



MYD-C7Z010-20-V2

Hardware User Manual

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History

Version	Author	Participants	Date	Description
V1.0	Sender		2022-04-20	Initial Version

1. Summarize

MYD-C7Z010-20-V2 is an Evaluation board development kit based on Xilinx XC7Z0XX series processor launched by Shen zhen MYIR Electronics Co., LTD. The Evaluation board is composed of Core board MYC-C7Z010-20-V2 and Baseboard MYB-C7Z010/20-I. This manual is convenient for users to understand the interface definition and function application of Evaluation board, and also has certain guiding significance for project development using MYC-C7Z010-20-V2, our company's Core board.

1.1. Introduction of the board

The Core board is designed as a 140pin connector connected to the Baseboard, which integrates XC7Z0XX series application processor, DDR3, NAND Flash, eMMC and other circuits in a space of 75x55mm.

The Baseboard adopts a double-sided device layout and is powered by 12V DC power. It provides abundant common ports for verifying various functions. And reserved a large number of expansion interfaces, customers can flexibly design and expand applications based on the Baseboard.

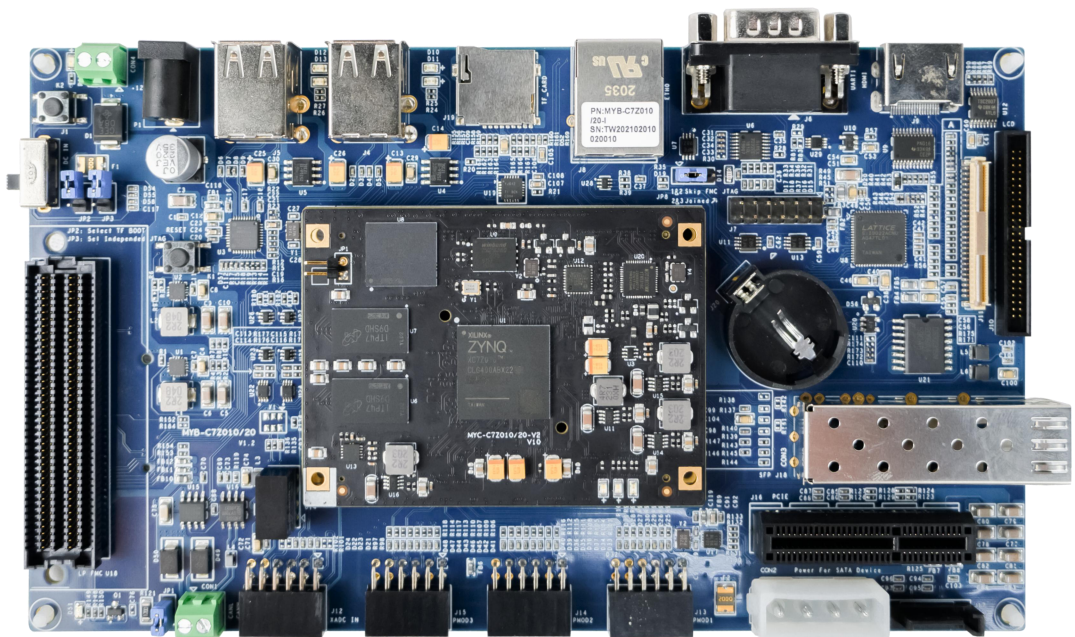


Figure 1-1 Appearance of MYD-C7Z010-20-V2

1.2. System chart

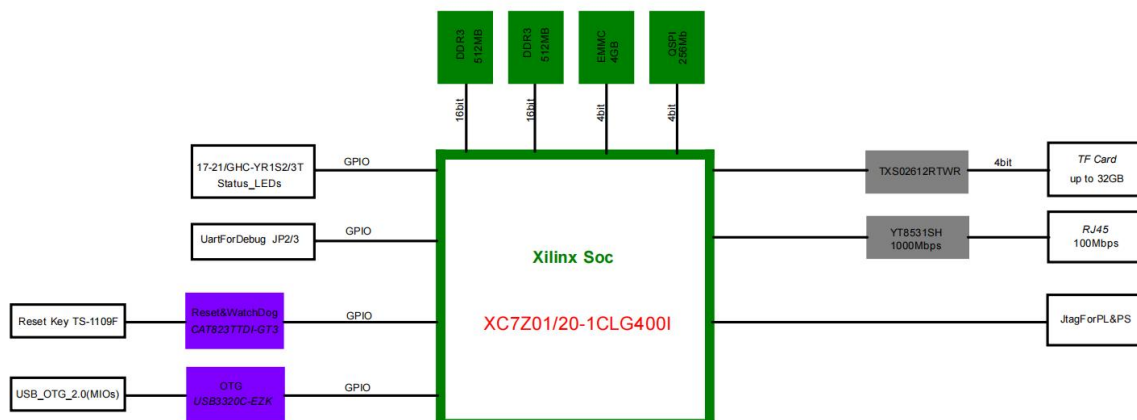


Figure 1-2 Core board system block diagram

1.3. Physical map

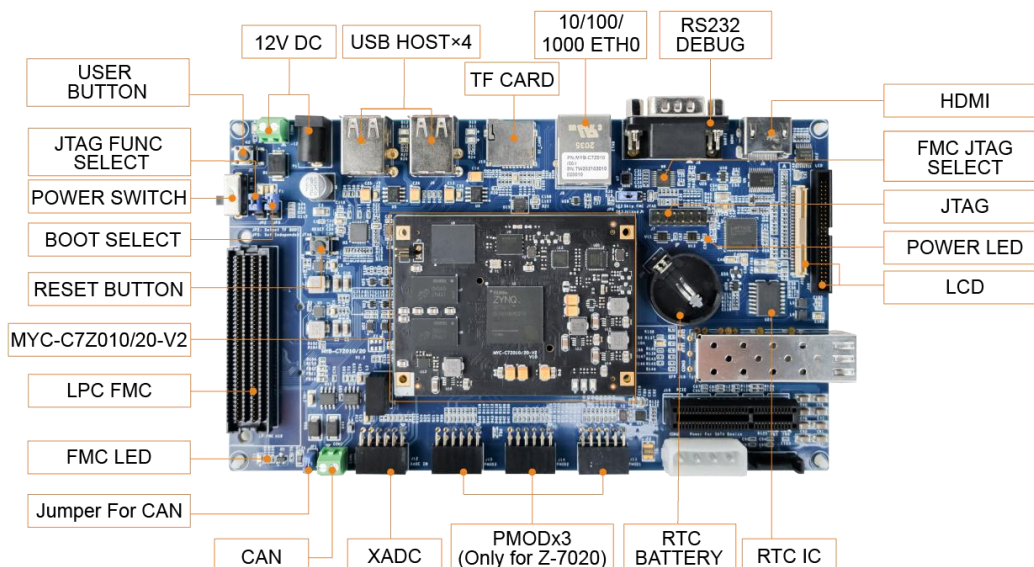


Figure 1-3 Labels on the front of the Evaluation board

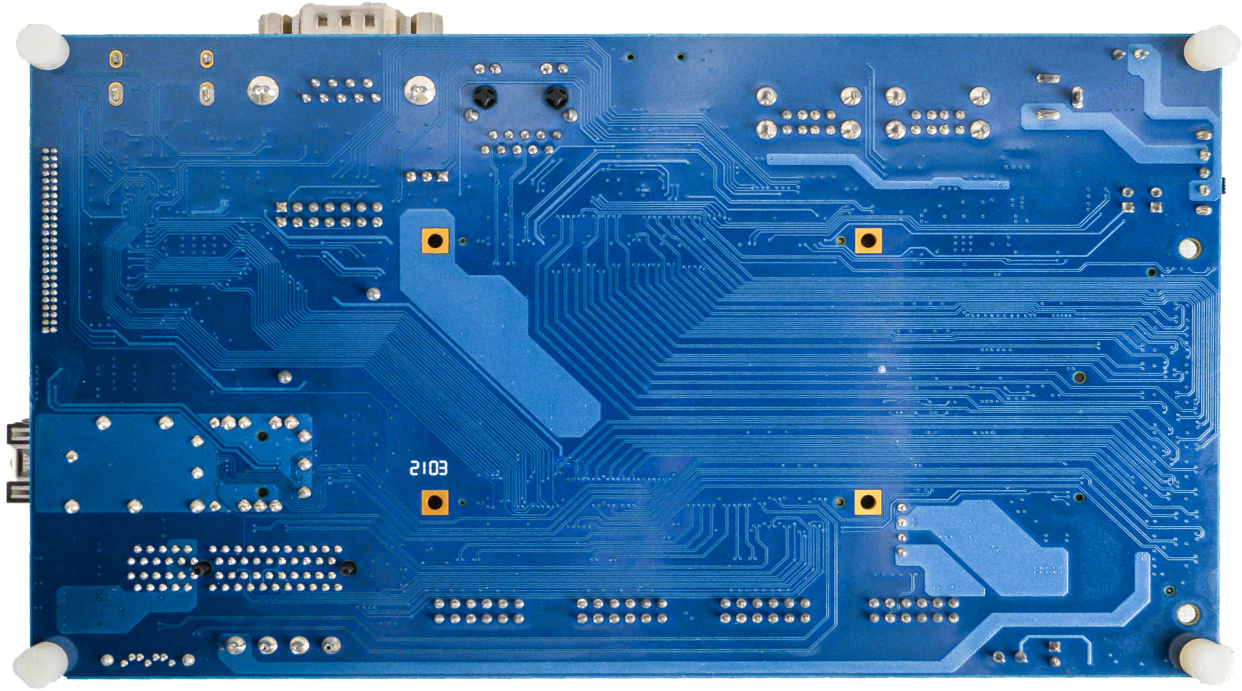


Figure 1-4 Label on the back of the Evaluation board

1.4. Main feature

Port	Describe
Power input	X1 12V DC 2.0mm power input
USB	X2 USB A port
Mode conversion row pin	X2 1x2Pin,2.54mm pin x1 1x2Pin,2.54mm CAN
Network	X1 1000M Ethernet RJ45 (Core board network port chip control)
CAN	x1 2Pin,3.5mm CAN x1 2Pin,3.5mm power port
Nine serial port	x1 RS-232 x1 RS-485 x1 CAN port
Expansion interface	x1 64P 1mm PCIE port x1 2x7Pin,2.54mm JTAG port
Micro SD	x1 Micro SD neck
Button	x1 Reset button x1 PS_BUTTON button
HDMI	x1 HDMI
SATA	x1 SATA

	x1 Power for SATA Device
FMC	X1 160Pin 1.27mm FMC
LCD	X1 2x25Pin,1.27mm LCD
XADC/PMOD1-3	x1 2x6Pin,2.54mm XADC x3 2x6Pin,2.54mm PMOD1-3
SFP	x1 SFP+ Card Edge SKT 20 POS 0.8mm
Light show	x1 Power light x1 RUN indicator light x1 FPGA indicator light

Table 1-1 Ports on the Baseboard

1.5. Reference data

MYIR Electronics provides supporting software and hardware materials, including but not limited to product manuals, hardware design guides, device manuals, software development guides, system images, etc., which can be downloaded at <http://down.myir-tech.com/>.

2. Power supply parameters

2.1. Power source tree

The system input voltage is 12V. The power path mainly supplies input voltage from 12V to 5V to the Core board. 5V mainly supplies power to the debugging serial port, CAN port, and power indicator circuits.

The Baseboard of the EVK Evaluation board and 5V of the Core board were powered on at the same time, and no abnormality was found after a large number of on-off tests. Because there are not many peripherals on EVK Evaluation board, the power on peripherals are relatively simple.

When designing a Baseboard, ensure that 5V Core boards are powered on the Baseboard first. For details, see the following circuit: 12V is the Baseboard input power, and VDD_5V is the Baseboard input power. 5V is converted to 3.3V on the Core board, and then power the bottom circuit at all places.

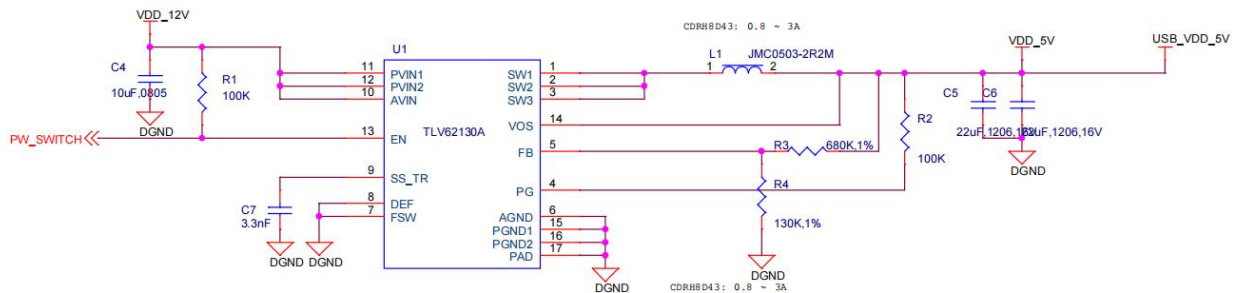


Figure 2-1 Shows the power-on time sequence of the Core board and Baseboard

2.2. Power consumption

Working conditions	Power supply voltage	Average current	Peak current	Total power consumption
Uboot boot process	12V	0.20A	0.25A	2.4W
Linux environment Do not connect peripherals	12V	0.23A	0.25A	2.76W
Linux environment Using one Ethernet	12V	0.26A	0.28A	3.12W

Table 2-1 Evaluating the power consumption of the board

Note: Power consumption test instrument main model: MEM DC Power Supply KXN-305D, 12V Power Supply.

2.3. Power Supply Requirements

MYD-Y7Z010-20-V2 Power supply design generally according to 12V input, Note do not use a power supply less than 12V but with a wide range of power supply. Ensure that the carrying capacity of the power supply is sufficient. That is, the total power consumed by the Evaluation board remains unchanged. When the supply voltage is lower, the output current of the power supply will be greater. If you have higher requirements for heat dissipation, you can use a radiator.

3. BOOT configurations

XC7Z0XX will first execute the chip internal curing Boot ROM when booting. Boot ROM starts by reading the Boot pin level state to enter different Boot sources. The two pins that determine the boot source are SD_BOOT_CONF_H and S_JTAG_H pins. These pins add a pull-up or pull-down design to the Core board. The Evaluation board is designed for two 2P JP2,JP3, and row pins. As shown in figure 3-1.

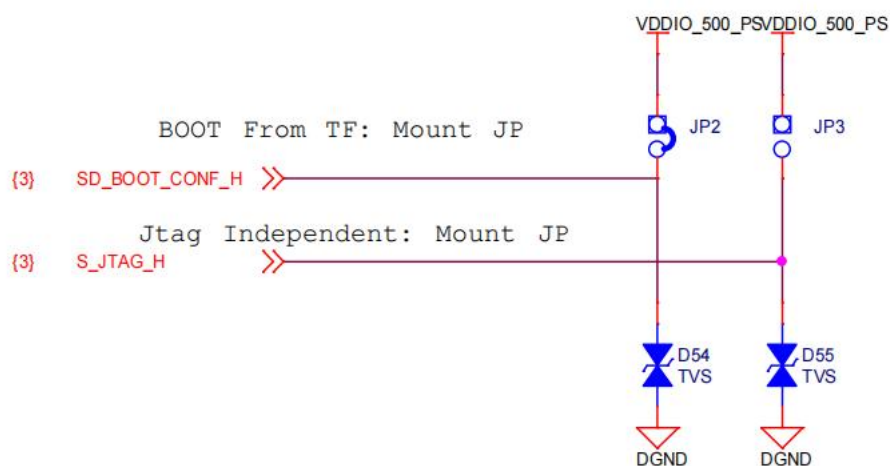


Figure 3-1 Enabling the DIP switch circuit

Table 3-1 lists the common boot sources.

Power switch		
SD_BOOT_CONF_H	S_JTAG_H	Instructions
0	0	Boot form JTAG
0	1	Default
1	1	Boot form Micro SD

Table 3-1 BOOT mode selection

4. Interface specification

The overall interface layout of the evaluation board is shown as follows, and the schematic diagram of subsequent partial interface circuits in this chapter corresponds to this layout.

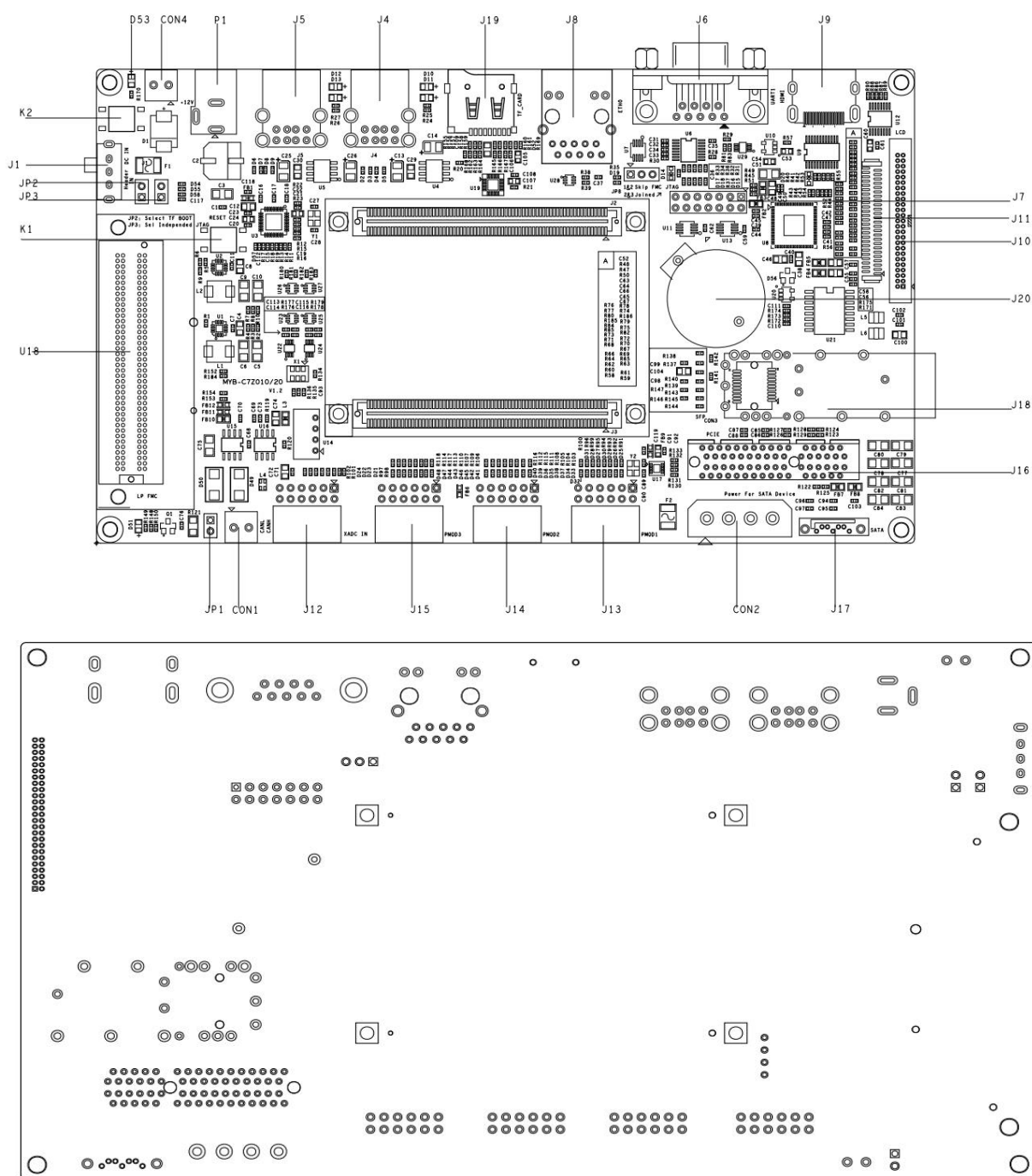


Figure 4-1 Ports on the MYD-C7Z010-20-V2 Evaluation board

4.1. Power interface

Power input connector: 49100500000W and uIO-TB11-381/3.5-02P-4010A. Use a 12V 2A DC adapter as the power input, and be sure not to use more than the specified operating voltage.

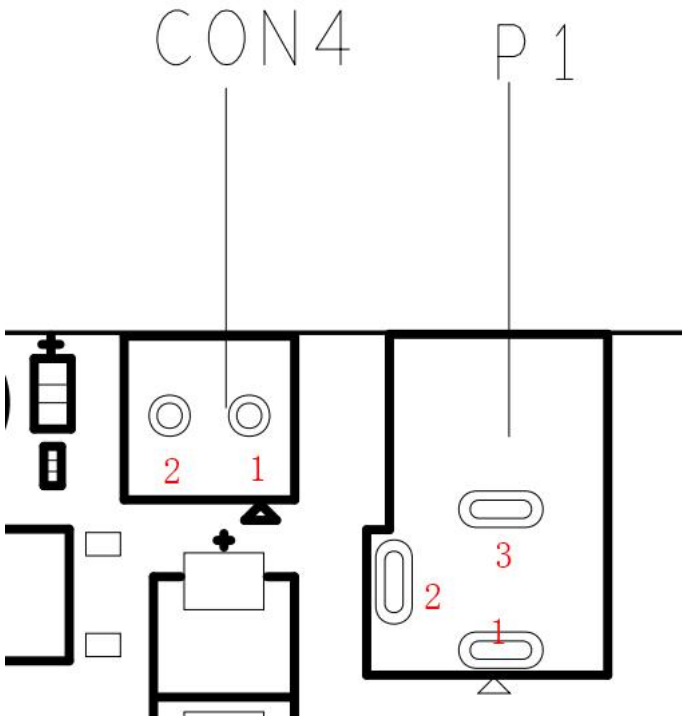


Figure 4-2 Schematic diagram of the power ports

4.1.1. Pin definition

Item	Identification	Function	Signal	Instructions
P1、 CON4	1	System power input, positive pole	12V	J1 and CON4 are both power supply interfaces, which can be connected to one other.
	2	System power input, negative pole	GND	
	3	System power input, negative pole	GND	

Table 4-1 Description of power ports

4.2. Debug serial port

4.2.1. RS232 debug serial port

The system debugging serial port uses J6 UART, MYD-C7Z010-20-V2 to design the UART interface, and the signal is RS232_R1IN, RS232_T1OUT, GND, and the signal pins are 2, 3 and 5. The debug serial port level is 3.3V IO.

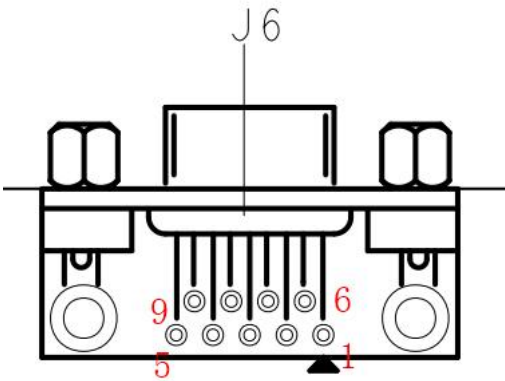


Figure 4-3 Debugging serial port

4.2.2. Pin definition

Item	Identification	Function	Signal	Instructions
J6	2	RS232 signal input	RS232_R1IN	
	3	RS232 signal output	RS232_T1OUT	
	5	signal gnd	GND	
	7、8			Short-circuiting through resistance

Table 4-2 Debugging serial port description

4.3. Power light

The Baseboard is designed with an LED lamp. D53 is the 3.3V power indicator light, which is blue when lit. The Core board is designed with 3 LED lights. D2 is the system running power indicator, which is blue when lit. D1 is FPGA indicator light, which is red when lit; D3 is the FMC indicator light, which is green when lit.

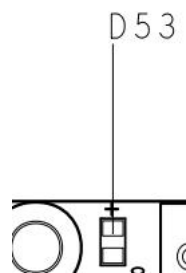


Figure 4-4 schematic diagram of LED indicators

4.3.1. Pin definition

Item	Identification	Function	Signal	Instructions
D53	POWER	Power indicator, blue	VDD_3.3V	On: The device is powered on properly Off: The device is powered off
D1	FPGA	FPGA indicator, red	VDD_3.3V	On: The device is powered on properly Off: The device is powered off
D2	POWER	Power indicator, blue	VDD_3.3V	On: The device is powered on properly Off: The device is powered off
D3	FMC	FMC indicator, green	VDD_3.3V	On: The device is powered on properly Off: The device is powered off

Table 4-3 Description of LED indicators

4.4. SD Card

A Micro SD card slot circuit is designed on the Evaluation board to support the operation of starting and swiping from Micro SD.

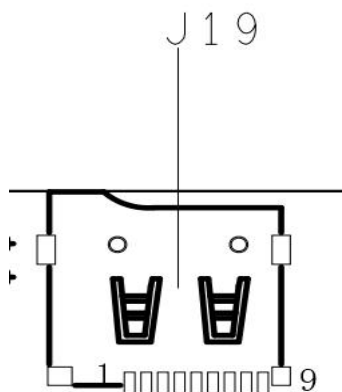


Figure 4-5 Micro SD card

4.4.1. Pin definition

Item	Identification	Function	Signal	Instructions
J19	1	Data bits 2	DAT2	
	2	Data bits 3	DAT3	
	3	order	CMD	
	4	Power 3.3 V	VCC	
	5	clock	CLK	
	6	Power ground	VSS	
	7	Data bits 0	DAT0	
	8	Data bits 1	DAT1	
	9	Card detection	CD	

Table 4-4 Ports on the micro SD card

4.4.2. Interface properties

Parameter	Test method	Minimum	Typical	Maximum	Unit	Description
writing speed	dd order	-	8.8	-	MB/s	time dd if=/dev/zero of=test_file bs=120M count=1 conv=fsync
reading speed	dd order	-	169	-	MB/s	time dd if=test_file of=/dev/null bs=120M count=1 conv=fsync

Table 4-5 Read/write speed tests of SD card interfaces

4.5. HDMI

The Evaluation board is designed with HDMI interface. High-definition multimedia interface is a kind of digital video/audio interface technology, which is a special digital interface suitable for image transmission. It can transmit audio and image signals at the same time, with the highest data transmission speed of 2.25GB/s, without digital/analog or analog/digital conversion before signal transmission.

The interface diagram is as follows:

HDMI and LCD do not share RGB signals, It is recommended to use them separate.

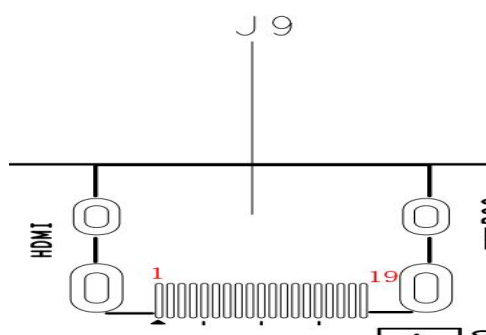


Figure 4-6 HDMI ports

4.5.1. Pin definition

Item	Identification	Function	Signal	Instructions
J9	1	HDMI data sends differential signals	D2+	
	3	HDMI data sends differential signals	D2-	
	4	HDMI data sends differential signals	D1+	
	6	HDMI data sends differential signals	D1-	
	7	HDMI data sends differential signals	D0+	
	9	HDMI data sends differential signals	D0-	
	10	HDMI clock positive signal	CLK+	
	12	HDMI clock negative signal	CLK-	
	13	Cec	CEC	
	15	clock signal	SCL	
	16	data signal	SDA	
	18	Power input	5V	
	19	Hot swap detection	HOTPLUG	
	2, 5, 8, 11, 17	Power ground/signal shielding cable	GND	

Table 4-6 HDMI ports description

4.6. LCD

The Evaluation board designs a 2x25_2.54mm double row pin interface, which defines multiple GPIO ports,LCD signals,FPGA signals, 3.3V,DC_IN_5V on this pin row. The interface figure is as follows.

HDMI and LCD do not share RGB signals, which may cause mutual interference. It is recommended to use them separately.

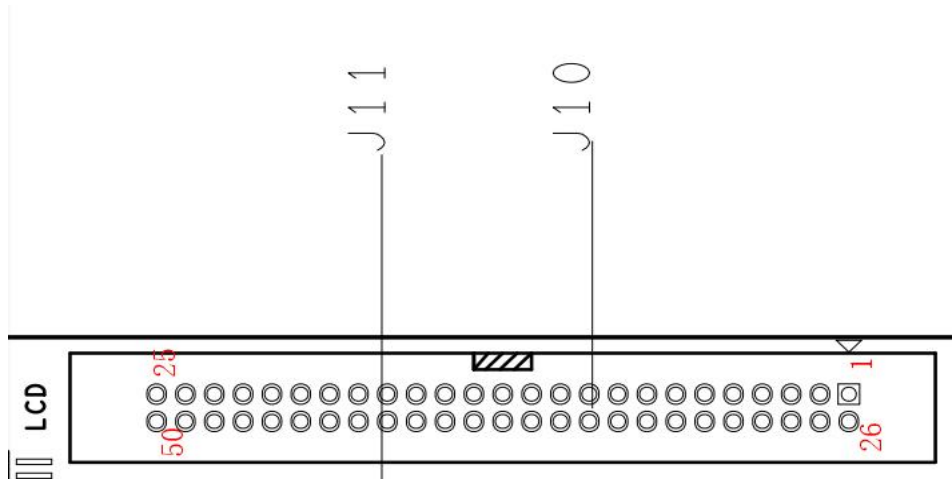


Figure 4-7 LCD expansion ports

4.6.1. Pin definition

Item	Identification	Function	Signal	Instructions
J10	1	Power 5V	VDD_5V	Power supply 5V input
	2	Power 5V	VDD_5V	Power supply 5V input
	3	Power 3.3V	VDD_3V3	Power supply 3.3V input
	4	Power 3.3V	VDD_3V3	Power supply 3.3V input
	5	Power gnd	DGND	
	6	Power gnd	DGND	
	7	Display data b0	b0	
	8	Display data b1	b1	
	9	Display data b2	b2	
	10	Display data b3	b3	
	11	Display data b4	b4	
	12	Display data b5	b5	
	13	Display data b6	b6	
	14	Display data b7	b7	
	15	Display data b8	b8	

	16	Display data b9	b9	
	17	Display data b10	b10	
	18	Display data b11	b11	
	19	Display data b12	b12	
	20	Display data b13	b13	
	21	Display data b14	b14	
	22	Display data b15	b15	
	23	Display data r0	r0	
	24	Display data r1	r1	
	25	Display data r2	r2	
	26	Display data r3	r3	
	27	Display data r4	r4	
	28	Display data r5	r5	
	29	Display data r6	r6	
	30	Display data r7	r7	
	31	Signal ground	DGND	
	32	Display backlight power PWM control	BL_CN	
	33	LCD reset signal	RESET	
	34	undefinition	DISP	
	35	Capacitive touch interrupt	TP_INTn	
	36	Capacitive touch reset	TP_RESET	
	37	I2C clock	TWC	
	38	I2C data	TWO	
	39	Signal ground	DGND	
	40	Enable display data	DEN	
	41	Display field signal	VSYNC	
	42	Display line signal	HYSNC	
	43	Display pixel clock signal	PCLK	
	44	Signal ground	DGND	
	45	undefinition	XR	
	46	undefinition	XL	
	47	undefinition	YD	
	48	undefinition	YU	
	49	undefinition	LR	
	50	Signal ground	DGND	

Table 4-7 LCD expansion ports description

4.7. USB

The Evaluation board is designed with A USB OTG interface, with a USB OTG circuit leading to the USB connector J4 and J5. USB_PHY_ID, set to USB HOST by default. When USB HOST interface and USB OTG interface are used as USB HOST function, VUBS pin will output 5V, which comes from the system power supply of the evaluation board. Therefore, ensure that the power provided to the Evaluation board is sufficient, otherwise the external device may not work or work improperly.

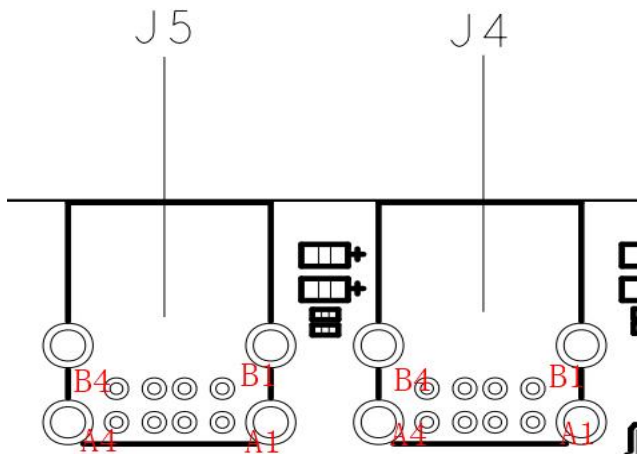


Figure 4-8 USB OTG port

4.7.1. Pin definition

Item	Identification	Function	Signal	Instructions
J4、J5	A1、B1	USB 5V power supply	5V_VBUS	Power supply 5V input
	A2、B2	USB OTG data negative	OTG1_USB_N	
	A3、B3	USB OTG data positive	OTG1_USB_P	
	A4、B4	Power ground	DGND	Signal ground
	5、6、7、8	Power ground	DGND	Fixed pin

Table 4-8 Description of USB host ports

4.7.2. Interface properties

Parameter	Test method	Minimum	Typical	Maximum	Unit	Description
Writing speed	dd order	-	8.8	-	MB/s	time dd if=/dev/zero of=test_file bs=120M count=1 conv=fsync

Table 4-9 Performance of USB OTG ports

4.8. Ethernet0

The Core board controls one 1000/100M Ethernet interface, using integrated network transformer integrated RJ45 connector. The signal of the network port connector J8 is from the ENET0 resource of RMI of the XC7Z0XX series.

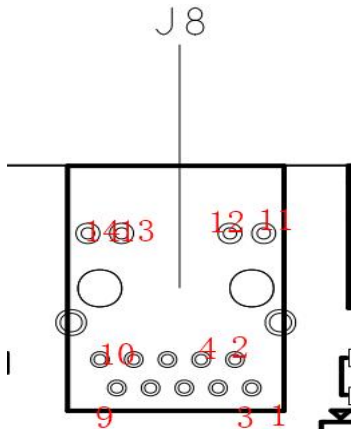


Figure 4-9 Ethernet interface signals

4.8.1. Pin definition

Item	Identification	Function	Signal	Instructions
J8	1	Transformer core	CT	
	2	Send and receive data 1 positive	TD1+	
	3	Send and receive data 1 minus	TD1-	
	4	Send and receive data 1 positive	RD1+	
	5	Send receive data 2 positive	TD2+	
	6	Send and receive data 2 minus	TD2-	
	7	Send and receive data 1 minus	RD1-	
	8	Send receive data 2 positive	RD2+	
	9	Send and receive data 2 minus	RD2-	
	10	Transformer center tap	CHS	
	11	Network connection Green positive indicator	GREEN+	
	12	Network connection Green indicator negative terminal	GREEN-	

	13	Network connection Positive yellow indicator	YELLOW+	
	14	Network connection Yellow indicator negative terminal	YELLOW-	

Table 4-10 Schematic diagram of Ethernet interface signals

4.8.2. Interface properties

Parameter	Test method	Minimum	Typical	Maximum	Unit	Description
TCP Network bandwidth	iperf3	-	924	-	Mb/s	server: iperf3 -s client: iperf3 -c 192.168.30.3 -i 2 -t 60
UDP Network bandwidth	iperf3	-	100	-	Mb/s	server: iperf3 -s client: iperf3 -c x.x.x.x -u -i 2 -t 60 -b 1G

Table 4-10 Schematic diagram of Ethernet interface signals

4.9. Button interface

The Evaluation board is designed with 2 buttons. One is K1, which is the power reset button, and one is K2, which is PS_BUTTON. Model TS-1109W, specification is 6*6*4.3mm.

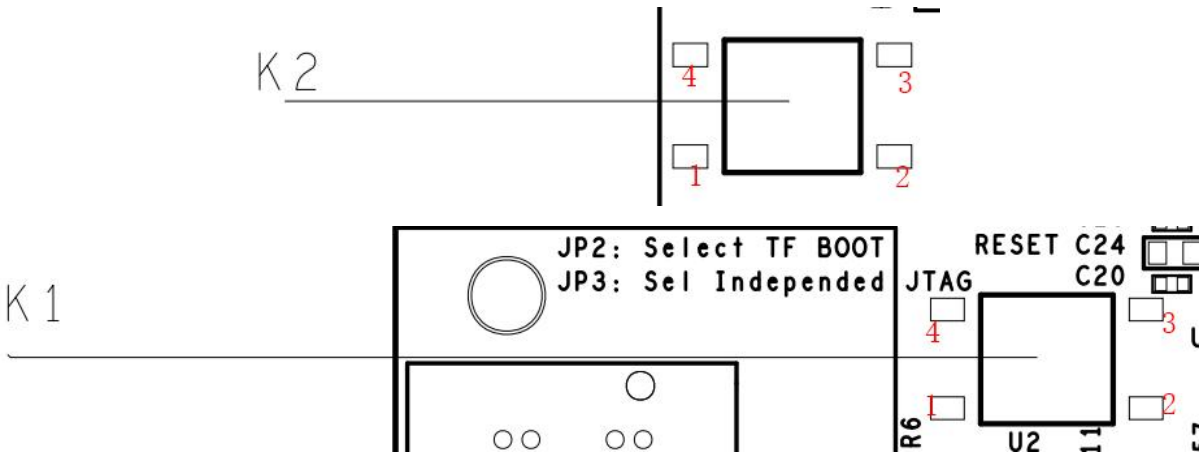


Figure 4-10 Schematic diagram of button signals

4.9.1. Pin definition

Item	Identification	Function	Signal	Instructions
K1	1、 2	Reset	EXT_RST_IN	Key pressing produces a reset
	3、 4	Power ground	DGND	
K2	1、 2	Power ground	1.8V	
	3、 4	Power ground	PS_BUTTON	PS button

Table 4-12 Button description

4.10. JTAG

The Evaluation board designs the JTAG interface, and the JTAG startup mode is considered to be a slave startup mode and is always an insecure startup mode. The JTAG chain can be configured in cascading and independent modes.

In standalone JTAG mode, users can only access Xilinx TAP through PLTDO/TMS/TCK/TDI. When users access the ARM DAP, they must use the PJTAG signal.



Figure 4-11 JTAG interface diagram

4.10.1. Pin definition

Item	Identification	Function	Signal	Instructions
J7	2	Power input	3.3V	
	4	JTAG mode selection	JTAG_TMS	
	6	JTAG signal clock	JTAG_TCK	
	8	JTAG signal output	JTAG_TDO	
	10	JTAG signal input	JTAG_TDI	
	14	JTAG reset signal	JTAG_NTRST	
	1, 3...13	Power ground	DGND	

Table 4-13 JTAG description

4.11. FMC

The Evaluation board designs the FMC interface, which provides standard sandwich board (sub card) sizes, connectors, and module interfaces for FPGAs on the Baseboard (carrier card). In this way, the I/O interface is separated from FPGA, which not only simplifies the I/O interface module design, but also maximizes the reuse of the load card. This model is ASP-134603-01. The specifications are Array Socket SKT 160Pin 1.27mm Solder. It has compatibility: standardized power supply, standardized signal definition, increased compatibility with each other.

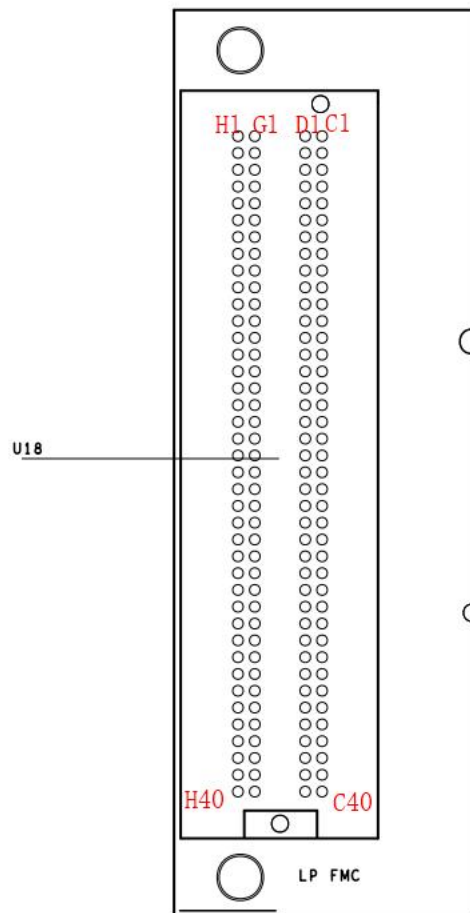


Figure 4-13 Ports on the FMC

4.11.1. Pin definition

Due to the excessive number of pins, I will not expand the description here, listing a few special instructions.

Item	Identification	Function	Signal	Instructions
U18	C35,C37	12V voltage from the mainboard to the sub-card	12P0V	
	C39	3.3V voltage from mainboard to sub-card	3P3V	
	D32	3.3V auxiliary voltage	3P3VAUX	
	H4	Differential clock signal from sub-card to motherboard	CLK[0..1]_M2C_N	
	H5	Differential clock signal from sub-card to motherboard	CLK[0..1]_M2C_P	
	G39	Adjustable voltage output from motherboard to subcard	VADJ	

Table 4-15 Ports on the FMC

4.12. CAN

CAN bus is designed in the Evaluation board. CAN bus is suitable for large-data short-distance communication or small-data long-distance communication, with high real-time requirements, multi-master, multi-slave or equal nodes. The interface diagram is as follows:

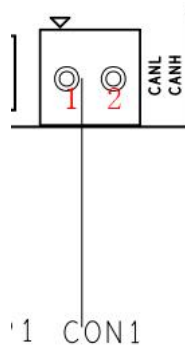


Figure 4-18 Schematics diagram of the CAN interface

4.12.1. Pin definition

Item	Identification	Function	Signal	Instructions
CON1	1	CAN low level	CANL	
	2	CAN high level	CANH	

Table 4-20 CAN ports description

4.13. RTC battery

The Evaluation board is designed with a button battery holder. The battery seat is 2Pinx1.25mm spacing. Button batteries with rated output voltage of 3.0V are required.

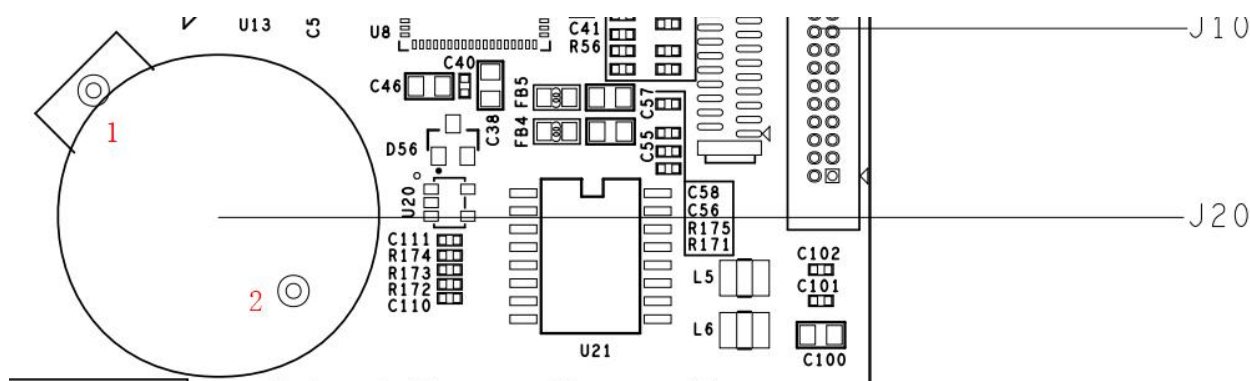


Figure 4-19 Schematics diagram of RTC ports

4.13.1. Pin definition

Item	Identification	Function	Signal	Instructions
J20	1	Button battery positive terminal	VDD_BAT	Button battery positive output
	2	DGND	DGND	Button battery negative

Table 4-21 Ports on the RTC battery

4.14. FPC connector

The Evaluation board is designed with FPC connectors. Connector model fh12-50s -0.5 SV, 50Pin,0.5mm,3.9mm Height. FPC has the characteristics of small size, light weight and thin thickness.

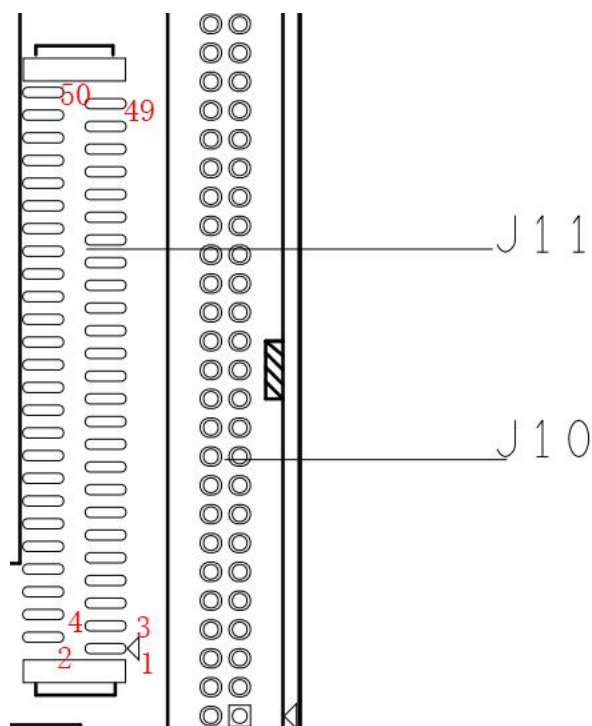


Figure 4-20 Ports on the FPC

4.14.1. Pin definition

Item	Identification	Function	Signal	Instructions
J11	1	Power 5V	VDD_5V	Power supply 5V input
	2	Power 5V	VDD_5V	Power supply 5V input
	3	Power 3.3V	VDD_3V3	Power supply 3.3V input
	4	Power 3.3V	VDD_3V3	Power supply 3.3V input
	5	Power gnd	DGND	
	6	Power gnd	DGND	
	7	Display data b0	b0	
	8	Display data b1	b1	
	9	Display data b2	b2	
	10	Display data b3	b3	
	11	Display data b4	b4	
	12	Display data b5	b5	
	13	Display data b6	b6	
	14	Display data b7	b7	
	15	Display data b8	b8	
	16	Display data b9	b9	
	17	Display data b10	b10	

18	Display data b11	b11	
19	Display data b12	b12	
20	Display data b13	b13	
21	Display data b14	b14	
22	Display data b15	b15	
23	Display data r0	r0	
24	Display data r1	r1	
25	Display data r2	r2	
26	Display data r3	r3	
27	Display data r4	r4	
28	Display data r5	r5	
29	Display data r6	r6	
30	Display data r7	r7	
31	Signal ground	DGND	
32	Display backlight power PWM control	BL_CN	
33	LCD reset signal	RESET	
34	undefinition	DISP	
35	Capacitive touch interrupt	TP_INTn	
36	Capacitive touch reset	TP_RESET	
37	I2C clock	TWC	
38	I2C data	TWO	
39	Signal ground	DGND	
40	Enable display data	DEN	
41	Display field signal	VSYNC	
42	Display line signal	HYSNC	
43	Display pixel clock signal	PCLK	
44	Signal ground	DGND	
45	undefinition	XR	
46	undefinition	XL	
47	undefinition	YD	
48	undefinition	YU	
49	undefinition	LR	
50	Signal ground	DGND	

Table 4-22 Ports on the FPC connector

5. Machine dimension

Evaluation plate technology: 190x110 X1.6mm, 6-layer plate, double-sided device layout, TG140 plate, gold sink process.

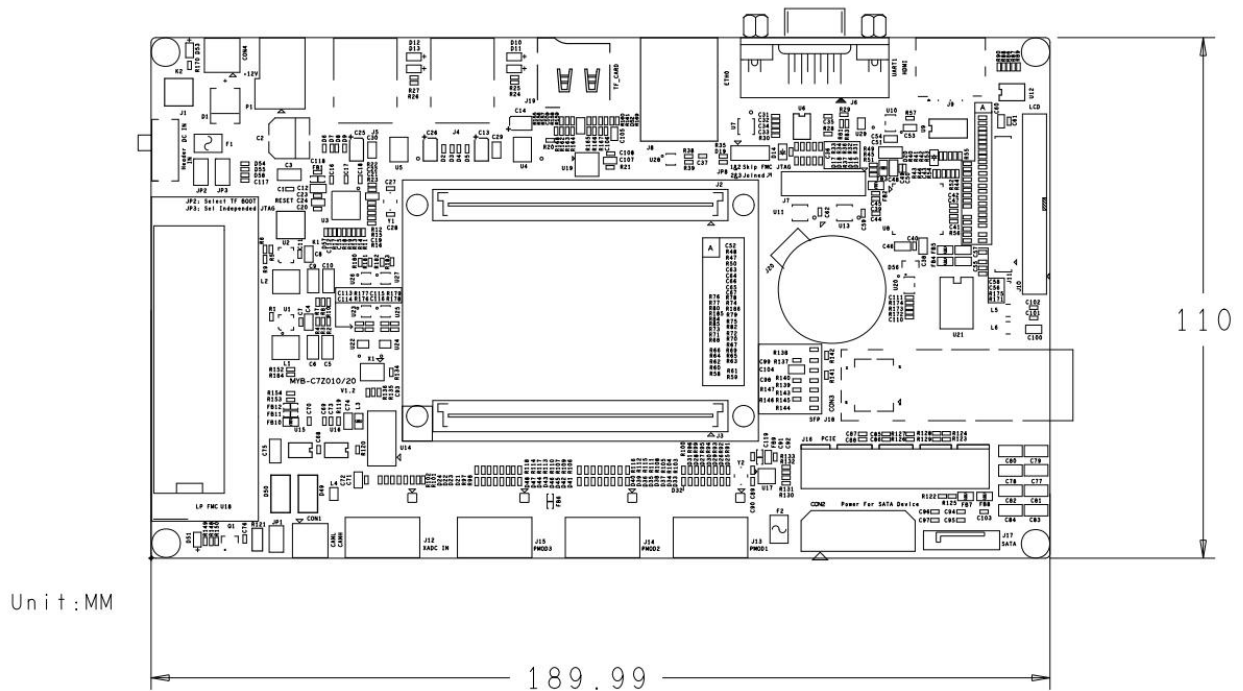


Figure 5-1 Evaluating the size of the board

6. Evaluate board kit configuration

6.1. Evaluation board product model

Note: XC7Z010 and XC7Z020 have the choice of 1CLG400I and 2CLG400I respectively.

Model	MYD-C7Z010-V2-4E1D-667-I	MYD-C7Z020-V2-4E1D-766-I
main chip	XC7Z010-1CLG400I	XC7Z020-2CLG400I
Working temperature	-40~+85°C	-40~+85°C
memory	512MB DDR3L	512MB DDR3L
storage	256Mb Flash	256Mb Flash
EMMC	4GB EMMC	4GB EMMC

Table 6-1 Optional evaluation board kits

6.2. Evaluation board package list

Item	Instructions
board card	x1 evaluation board
data	x1 Quick use manual
wire	x1 serial port x1 DC converter head x1 12V@2A Switching power adapter

Table 6-2 Packing list

7. Evaluate board connector definition

Item	Material model	brand	Item number
Power interface	49100500000W	LEYCONN	P1
CAN、Power interface	ULO-TB11-381/3.5-02P-4010A	ULO	CON1、CON4
Power for SATA	62012041130J	MOLEX	CON2
SFP seat	MECT-110-01-M-D-RA1	Leyconn	CON3
Power switch	SK-12F14	TOY	J1
Connector	61083-141402LF	FCI	J2、J3
USB port	4312821A03WH	Leyconn	J4、J5
DB9	50209132110H	Leyconn	J6
Double row needle	218206253010	Leyconn	J7
Network interface	HY911130AE	Hanrun	J8
HDMI port	44020411232W	Leyconn	J9
LCD interface	303500135130	Leyconn	J10
FPC	FH12-50S-0.5SV	HRS	J11
XADC、PMOD	218206253010	Leyconn	J12、J13、J14、J15
PCIE	2EG03217-D2D-DF	Foxconn	J16
SATA	067491-0020	MOLEX	J17
SFP	74737-0009	MOLEX	J18
Micro SD	MR01A-01211	ATOM	J19
RTC battery holder	46902120001J	Leyconn	J20
Reset button	TS-1109W	BEST	K1、K2
Single needle	3131025311161	Leyconn	JP1、JP2、JP3、JP8
PCI connector	2EG03217-D2D-DF	Foxconn	U18
LED Light	17-215/BHC-AP1Q2/3T	Everlight	D51、D53

Table 7-1 Materials and models of Baseboard connectors

8. Appendix A

8.1. Warranty & Technical Support Services

MYIR Electronics Limited is a global provider of ARM hardware and software tools, design solutions for embedded applications. We support our customers in a wide range of services to accelerate your time to market.

MYIR is an ARM Connected Community Member and work closely with ARM and many semiconductor vendors. We sell products ranging from board level products such as development boards, single board computers and CPU modules to help with your evaluation, prototype, and system integration or creating your own applications. Our products are used widely in industrial control, medical devices, consumer electronic, telecommunication systems, Human Machine Interface (HMI) and more other embedded applications. MYIR has an experienced team and provides custom design services based on ARM processors to help customers make your idea a reality.

The contents below introduce to customers the warranty and technical support services provided by MYIR as well as the matters needing attention in using MYIR' s products.

Service Guarantee

MYIR regards the product quality as the life of an enterprise. We strictly check and control the Core board design, the procurement of components, production control, product testing, packaging, shipping and other aspects and strive to provide products with best quality to customers. We believe that only quality products and excellent services can ensure the long-term cooperation and mutual benefit.

Price

MYIR insists on providing customers with the most valuable products. We do not pursue excess profits which we think only for short-time cooperation. Instead, we hope to establish long-term cooperation and win-win business with customers. So we will offer reasonable prices in the hope of making the business greater with the customers together hand in hand.

Delivery Time

MYIR will always keep a certain stock for its regular products. If your order quantity is less than the amount of inventory, the delivery time would be within three days; if your order quantity is greater than the number of inventory, the delivery time would be always four to six weeks. If for any urgent delivery, we can negotiate with customer and try to supply the goods in advance.

Technical Support

MYIR has a professional technical support team. Customer can contact us by email (support@myirtech.com), we will try to reply you within 48 hours. For mass production and customized products, we will specify person to follow the case and ensure the smooth production.

After-sale Service

MYIR offers one year free technical support and after-sales maintenance service from the purchase date. The service covers:

Technical support service

MYIR offers technical support for the hardware and software materials which have provided to customers;

- To help customers compile and run the source code we offer;
- To help customers solve problems occurred during operations if users follow the user manual documents;
- To judge whether the failure exists;
- To provide free software upgrading service.
- However, the following situations are not included in the scope of our free technical support service:
 - Hardware or software problems occurred during customers' own development;
 - Problems occurred when customers compile or run the OS which is tailored by themselves;
 - Problems occurred during customers' own applications development;

- Problems occurred during the modification of MYIR' s software source code.

After-sales maintenance service

The products except LCD, which are not used properly, will take the twelve months free maintenance service since the purchase date. But following situations are not included in the scope of our free maintenance service:

- The warranty period is expired;
- The customer cannot provide proof-of-purchase or the product has no serial number;
- The customer has not followed the instruction of the manual which has caused the damage the product;
- Due to the natural disasters (unexpected matters), or natural attrition of the components, or unexpected matters leads the defects of appearance/function;
- Due to the power supply, bump, leaking of the roof, pets, moist, impurities into the boards, all those reasons which have caused the damage of the products or defects of appearance;
- Due to unauthorized weld or dismantle parts or repair the products which has caused the damage of the products or defects of appearance;
- Due to unauthorized installation of the software, system or incorrect configuration or computer virus which has caused the damage of products.

Warm tips:

- MYIR does not supply maintenance service to LCD. We suggest the customer first check the LCD when receiving the goods. In case the LCD cannot run or no display, customer should contact MYIR within 7 business days from the moment get the goods.
- Please do not use finger nails or hard sharp object to touch the surface of the LCD.
- MYIR suggests user purchasing a piece of special wiper to wipe the LCD after long time use, please avoid clean the surface with fingers or hands to leave fingerprint.

- Do not clean the surface of the screen with chemicals.
- Please read through the product user manual before you using MYIR ' s products.
- For any maintenance service, customers should communicate with MYIR to confirm the issue first. MYIR' s support team will judge the failure to see if the goods need to be returned for repair service, we will issue you RMA number for return maintenance service after confirmation.

Maintenance period and charges

MYIR will test the products within three days after receipt of the returned goods and inform customer the testing result. Then we will arrange shipment within one week for the repaired goods to the customer. For any special failure, we will negotiate with customers to confirm the maintenance period.

For products within warranty period and caused by quality problem, MYIR offers free maintenance service; for products within warranty period but out of free maintenance service scope, MYIR provides maintenance service but shall charge some basic material cost; for products out of warranty period, MYIR provides maintenance service but shall charge some basic material cost and handling fee.

Shipping cost

During the warranty period, the shipping cost which delivered to MYIR should be responsible by user; MYIR will pay for the return shipping cost to users when the product is repaired. If the warranty period is expired, all the shipping cost will be responsible by users.

Products Life Cycle

MYIR will always select mainstream chips for our design, thus to ensure at least ten years continuous supply; if meeting some main chip stopping production, we will inform customers in time and assist customers with products updating and upgrading.

Value-added Services

- MYIR provides services of driver development base on MYIR' s products, like serial port, USB, Ethernet, LCD, etc.
- MYIR provides the services of OS porting, BSP drivers' development, API software development, etc.
- MYIR provides other products supporting services like power adapter, LCD panel, etc.
- ODM/OEM services.

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